

Machine Learning Interview Questions and Answers

Interview-focused revision guide with model answers.

Q1. What is Machine Learning?

Answer: Machine Learning is a subset of AI that enables systems to learn patterns from data and make predictions without being explicitly programmed.

Q2. Difference between AI, ML, and Deep Learning?

Answer: AI is the broader field. ML is a subset of AI that learns from data. Deep Learning is a subset of ML that uses neural networks with multiple layers.

Q3. What is Overfitting?

Answer: Overfitting occurs when a model learns noise in the training data and performs poorly on unseen data.

Q4. What is Underfitting?

Answer: Underfitting occurs when a model is too simple to capture the underlying patterns in the data.

Q5. What is the Bias-Variance Tradeoff?

Answer: Bias is error from overly simplistic assumptions. Variance is error due to sensitivity to training data. Good models balance both.

Q6. What is Cross Validation?

Answer: A technique to evaluate model performance by splitting data into multiple folds and training/testing repeatedly.

Q7. What is Feature Engineering?

Answer: The process of creating, transforming, or selecting variables to improve model performance.

Q8. Difference between Classification and Regression?

Answer: Classification predicts categories, while regression predicts continuous numerical values.

Q9. What is Logistic Regression?

Answer: A classification algorithm that uses the sigmoid function to predict probabilities.

Q10. What is Random Forest?

Answer: An ensemble method that combines multiple decision trees using bagging to improve accuracy and reduce overfitting.

Q11. What is XGBoost?

Answer: A gradient boosting algorithm known for high performance, regularization, and efficient handling of large datasets.

Q12. What is PCA?

Answer: Principal Component Analysis reduces dimensionality while preserving as much variance as possible.

Q13. What is Precision and Recall?

Answer: Precision measures correctness of positive predictions. Recall measures how many actual positives are identified.

Q14. What is ROC-AUC?

Answer: A metric that evaluates classification performance across different decision thresholds.

Q15. What is Regularization?

Answer: A technique to reduce overfitting by penalizing large model coefficients.

Python Coding Interview Questions

Q1. How do you split data into train and test sets?

Answer: Use `train_test_split` from `sklearn.model_selection`.

Q2. How do you handle missing values?

Answer: Use mean/median/mode imputation or model-based approaches.

Q3. How do you evaluate a classification model?

Answer: Use confusion matrix, accuracy, precision, recall, F1-score, and ROC-AUC.

Q4. How do you evaluate a regression model?

Answer: Use MAE, MSE, RMSE, and R^2 score.